

1. Takeover Event

Robot heads toward obstacle; operator grabs joystick and steers sharp right.

- Pre-crash drift: $[s_{t-k}, a_{\text{drift}}]$
- Human panic save: $[s_t, a_{\text{steer}}]$
- Closed-loop endogeneity unmanaged

2. Naive Logging

Single contiguous stream recorded without takeover boundary tag.

- Drift and save merged blindly
- No demarcation of human input
- Covariate shift enters dataset

3. Policy Collapse

Retrained policy fits spurious causal link between obstacle proximity and turn.

- **Spurious Rule:** "Speed into pillars"
- Multi-modal distribution blur
- Catastrophic bench collision

THE GOVERNED DATA FILTER BOUNDARY

1. Hardware Tagging

Real-time MCU detects takeover $\|\mathbf{u}_h\| > \delta_{\text{takeover}}$ and stamps PTP hardware clock.

- Takeover timestamp t_{takeover}
- IEEE 1588 PTP sync ($\pm 1 \mu\text{s}$)
- Hardware telemetry lineage nonce

2. Episode Slicing

Automated partition isolates pre-crash failure from corrective recovery.

- Discard window: $[-T_{\text{discard}}, 0] \rightarrow \mathcal{D}_{\text{fail}}$
- Recovery demo: $[0, T_{\text{rec}}] \rightarrow \mathcal{D}_{\text{recov}}$
- Nominal slice: $\mathcal{D}_{\text{nominal}}$ pristine

3. Governed Lineage

Cryptographic hash chain binds trained weights to audited dataset partition.

- $\text{SHA-256}(\mathcal{D}_{\text{clean}}) \leftrightarrow \text{SHA-256}(W)$
- Shadow-mode replay evaluation gate
- Stable autonomous release

DOSSIER AUDIT TRAIL (AUTH-01): $\text{dataset_sha256} = 8f7b3a\dots \longleftrightarrow$
 $\text{model_weights_sha256} = e3b0c4\dots \longleftrightarrow \text{active_bank} = \text{BANK_A}$